

Curating Network Data And Descriptive Statistics

Empirical Methods — Week 18

Teaching deck draft

Week 18 question

- How should researchers define, measure, and summarize relationships that shape economic outcomes?
- Network curation comes before network causal inference.
- Node, edge, weight, direction, timing, and boundary choices shape the estimand.
- A graph is not evidence until the link meaning is justified and auditable.

This is the opening lecture of the optional network-methods block.

Nodes, edges, and bipartite data

$$A_{ij} = \begin{cases} 1 & \text{if there is a link from } i \text{ to } j, \\ 0 & \text{otherwise.} \end{cases} \quad B_{if} = \begin{cases} 1 & \text{if worker } i \text{ is linked to firm } f, \\ 0 & \text{otherwise.} \end{cases}$$

$$P_{ij}^{worker} = \sum_f B_{if} B_{jf}$$

- Nodes may be workers, firms, schools, managers, households, vacancies, or neighborhoods.
- Edges may be referrals, coworking, co-residence, communication, assignment, or transactions.
- Direction and weight matter when recommendations, supervision, duration, or intensity are asymmetric.
- Bipartite data are often the raw object; projection creates a new empirical object.

Network definitions are substantive

Choice	Empirical version	Economic content
Node	worker, firm, vacancy, household	whose choices and outcomes matter
Edge	referral, coworker, neighbor, message	information, trust, exposure, monitoring
Direction	$i \rightarrow j, j \rightarrow i$, undirected	asymmetric recommendation or influence
Weight	duration, frequency, value	strength of relationship or exposure
Timing	month, spell, year, cohort	whether exposure precedes outcome
Boundary	firm, city, platform, survey frame	which links are unobserved or truncated

The same raw records can imply different network variables under different mechanisms.

- Referrals: who hears about vacancies and who is recommended.
- Information transmission: how job knowledge travels through contacts, neighborhoods, and coworkers.
- Coworker exposure: mentoring, norms, monitoring, team learning, and job ladders.
- Neighborhood job networks: local residents connected to employers and opportunities.
- Learning and diffusion: practices, technologies, beliefs, and search strategies.
- Bargaining and inequality: unequal access to high-value contacts and outside options.

Anchor papers: Autor; Bayer, Ross, and Topa; Beaman and Magruder; Barwick et al.; Hellerstein et al.

Data sources and construction

Source	Natural graph	Main limitation
Linked employer–employee	worker–firm bipartite; coworker projection	informal ties outside firms
Referral-program records	directed employee–applicant links	only formal referrals
Survey rosters	friendship, kinship, advice, support	recall error and missing weak ties
Platform logs	worker–client or interaction graph	platform rules and proprietary access
School or cohort rosters	peer exposure graph	assignment and co-presence are not random
Neighborhood records	local opportunity or co-residence graph	proximity is not interaction

Preserve the raw relational object before computing projected links or summaries.

Descriptive statistics are theory-laden

$$d_i = \sum_j A_{ij} \quad s_i = \sum_j W_{ij} \quad c_i = \lambda^{-1} \sum_j A_{ij} c_j$$

$$H = \frac{\sum_i \sum_j A_{ij} \mathbf{1}\{g_i = g_j\}}{\sum_i \sum_j A_{ij}}$$

- Degree can mean opportunity access, referral reach, survey visibility, or firm size.
- Strength adds intensity but can hide whether exposure is broad or concentrated.
- Centrality can proxy information access only when paths carry the relevant information.
- Homophily describes similarity among linked nodes; it does not separate sorting from influence.

Pitfalls and diagnostics

Risk	Diagnostic
Missing links	distinguish unobserved ties from true zero ties
Endogenous formation	state whether the network is an outcome, exposure, or design object
Boundary truncation	test alternative graph frames and inspect isolates/components
Projection error	compare bipartite statistics with projected network statistics
Homophily vs influence	avoid causal language until formation and interference are handled
Privacy	report how masking changes degree, strength, and components
Dynamic mismatch	vary time windows and check exposure precedes outcome

Practical rules

- ① State the labor-market mechanism before defining the graph.
- ② Define node, edge, direction, weight, layer, timing, and boundary separately.
- ③ Preserve raw edge lists, bipartite spell tables, and survey rosters.
- ④ Build the primary network statistic and one plausible alternative.
- ⑤ Report missing-link risk and privacy masking as measurement issues.
- ⑥ Normalize projections when institution size mechanically creates links.
- ⑦ Test whether patterns are driven by hubs, components, or boundary truncation.
- ⑧ Keep descriptive diagnostics separate from causal robustness.

- Autor: labor markets are wired through search channels and intermediaries.
- Bayer, Ross, Topa: residential proximity can reveal informal hiring networks.
- Beaman, Magruder: referral incentives affect worker selection.
- Friebel et al.: employee referral programs as a management practice.
- Barwick et al.: referrals can shape labor-market inequality.
- Hellerstein, McInerney, Neumark: networks matter after mass layoffs.

The lesson is not “compute centrality.” The lesson is: define the economic relation.

Reproduce

- synthetic workers
- neighborhoods
- employers
- bipartite links
- local exposure

Diagnose

- boundary choices
- edge rules
- binary vs weighted
- missing links
- projection risk

Transfer

- directed referrals
- in-degree/out-degree
- referral success
- inequality memo
- causal preview

Where curation stops

- Curation defines edge lists, adjacency matrices, bipartite objects, and summaries.
- Curation can diagnose measurement sensitivity, missing links, and graph boundaries.
- It does not solve reflection, sorting, peer effects, or interference.
- Network causal inference begins when the researcher claims counterfactual effects through links.
- Lecture 19 takes up exposure mappings, peer effects, network experiments, and dependence.

Takeaways

- Network definition is part of the estimand.
- A link is an economic channel, not just a row in an edge list.
- Bipartite and multilayer objects should not be collapsed without a reason.
- Degree, centrality, clustering, and homophily are descriptive until the mechanism is defended.
- Missing links, boundaries, privacy, and timing can change the object.
- Strong curation makes Lecture 19's causal designs possible.